

- 8 Using the enthalpy changes below, calculate the standard enthalpy change of formation of gaseous hydrogen chloride.

	enthalpy change/ kJ mol^{-1}
$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$	-92
$\text{N}_2(\text{g}) + 4\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{NH}_4\text{Cl}(\text{s})$	-629
$\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$	-176

- A** $-46.3 \text{ kJ mol}^{-1}$
- B** $-92.5 \text{ kJ mol}^{-1}$
- C** -180 kJ mol^{-1}
- D** -361 kJ mol^{-1}